Ctrl+Z Burnout

Academic Decision Simulator

Milestone 2: Users and Data:

What Will We Cover?

- **3 Fictional Personas:** Diverse university students representing varying technical levels and accessibility needs.
- **Bias-Check Notes:** Our process for avoiding stereotypes and ensuring inclusive design.
- **Data Source (SUCWD):** The synthetic dataset powering our workload simulations.

Part 1: Fictional Personas

Persona 1: Samuel Okafor, the Returning Working Student

- **Name:** Samuel Okafor, 38
- **Background:** First-generation university student returning to study after 15 years in warehouse logistics. He is a part-time Business Administration student and works rotating shifts as a shift supervisor. He has two children. His shifts change every two weeks, so his free hours are never the same twice. He pays tuition out of pocket, one semester at a time.
- **Tech Level:** Below typical. He is comfortable with his phone and WhatsApp, but finds university portals confusing. He avoids multi-step web tools.
- **AI Familiarity:** Novice. He has heard of chatbots but has never used one for anything important. He is unsure whether to trust one.
- **Accessibility Need:** Dyslexia. Dense paragraphs, small fonts and long course-code lists slow him down and cause errors. He works best with short sentences, clear spacing, numbers instead of long text, and read-aloud support.
- **Goals:**
    - Finish his degree in a realistic number of years without quitting his job.
    - See clearly what a course will “cost” in weekly hours before he pays for it.
    - Choose a semester load that fits his shift pattern.
- **Pain Points:**
    - Course descriptions are long and written in academic language.
    - Advisors have little time, and he cannot attend office hours during the day.
    - He has taken on too many courses before and had to withdraw, losing money.
    - He distrusts tools that “decide for him” and wants to see the reasoning behind a suggestion.

Persona 2: Lucía Fernández, the Commuting First-Year

- **Name:** Lucía Fernández, 19
- **Background:** Second-semester Biology student who lives with her family in a town 90 minutes away by bus. She helps run the family shop on weekends. Her commute eats about three hours of her day, so she studies on the bus with a weak signal. She shares one laptop with her younger brother.
- **Tech Level:** Typical. She is comfortable with apps, spreadsheets and the learning management system (LMS).
- **AI Familiarity:** Novice. She has tried a chatbot once for a homework question and found the answer too long and too generic.
- **Accessibility Need:** ADHD (combined type). She struggles with time estimation and with schedules that show everything at once. Cluttered calendars overwhelm her, and she tends to freeze or over-commit. She does best with one decision at a time, visible progress, and gentle reminders she can adjust.
- **Goals:**
    - Build a first-year schedule that she will actually follow.
    - Understand how heavy a course combination really is, including commute and shop hours.
    - Test “what if” choices (drop a lab, add an elective) without making them final.
- **Pain Points:**
    - She underestimates assignment time and then crunches before deadlines.
    - Registration opens at a set time, and she feels pressure to decide quickly.
    - Planning tools show far too many options and settings at once.
    - She feels guilty about missed plans, and tools that nag or flag “failure” make this worse.

Persona 3: Nour Haddad, the International CS Junior

- **Name:** Nour Haddad, 22
- **Background:** Computer Science junior from Jordan studying abroad on a partial scholarship. She must keep a full-time course load and a minimum GPA to retain funding. English is her third language. She tutors part-time for living costs, and she plans each term carefully because a failed course would affect her funding and her residence permit.
- **Tech Level:** Advanced. She writes code, uses scripting to organize her own schedule, and reads documentation by default.

- **AI Familiarity:** Experienced. She uses large language models daily, and is skeptical of them. She notices when a tool hides its assumptions or states guesses as facts.
- **Accessibility Need:** None formally. She prefers plain, literal English without idioms, and benefits from tools that can be set to simple language.
- **Goals:**
    - Optimize her remaining semesters for prerequisite order, GPA protection and graduation timing.
    - Inspect and edit the simulator's assumptions (for example, her own estimate of weekly hours for a course).
    - Compare several plans side by side and export them.
- **Pain Points:**
    - Generic advice ignores her visa, scholarship and funding constraints.
    - Opaque recommendations: she wants to know why a tool suggests something.
    - Rigid tools that cannot be overridden or customized.
    - Low-quality workload data (for example, "average 10 hours") with no variation shown.

Part 2: Bias-Check Notes

Process. After drafting, we reviewed each persona against a checklist of common persona stereotypes, asked of each one: "Would this person still make sense if we swapped their gender, culture or ability?" We then revised any persona where the answer was no. We also checked the set as a whole for balance (who is tech-savvy, who is struggling, whose circumstances are treated as problems).

Stereotype or exclusion reviewed	Risk in a first draft	What we changed
Gender roles in tech	Default pattern: male = advanced, female = novice or "needs help."	Nour, a woman, is the advanced and skeptical user. Samuel, a man, is the below-typical user and openly unsure of himself. Lucía, a woman, is typical rather than "helpless."

Stereotype or exclusion reviewed	Risk in a first draft	What we changed
Neurodivergence as a burden	ADHD or dyslexia framed as the cause of failure, or as something the tool must “fix.”	Both needs are described as interaction preferences (one decision at a time, short text, read-aloud) rather than deficits. The persona’s problems come from poorly designed schedules and tools, not from the person. Neither persona is defined only by the condition.
Unlimited free time	Assuming students can study any evening or weekend.	Samuel has shifting shifts and children, Lucía has a 3-hour commute and weekend shop work, and Nour tutors part-time. Time scarcity is part of every persona’s situation.
Assuming “typical” access	Everyone has a private laptop, fast internet and a quiet place.	Lucía shares a laptop and has weak signal on the bus. We made the design implication explicit: the simulator should work on a phone and with low bandwidth.
Socio-economic assumptions	Assuming families pay tuition, or that dropping a course is cheap.	Samuel pays out of pocket and loses money on withdrawals. Nour’s funding and residence status depend on credits and GPA.
Cultural and language stereotypes	International student = “poor English” or “needs hand-holding”; stereotyped names and traits.	Nour is a high-performing user whose language need is a simple preference, not a deficit. Names and backgrounds vary without tying a trait to a culture.

Stereotype or exclusion reviewed	Risk in a first draft	What we changed
Age stereotypes	"Older students are bad with technology."	Samuel's low tech comfort comes from limited experience with university systems, not his age, and he is comfortable with phone tools. His age is not presented as a reason for the gap.
Single-story personas	Each person has only one problem (disability or poverty or language).	Each persona has several overlapping circumstances, plus real strengths (Samuel's planning discipline from work, Lucía's multitasking, Nour's technical skill).
Over-representing the typical user	Designing only for the "average" student.	Two of three personas are novices, and the set spans three tech levels, so the design is tested at both ends.

Remaining limits. Three personas cannot represent all students. Disabilities such as blindness, motor impairments or deafness, and students with caring duties or in different countries' university systems, are not covered. We recommend real-user testing with students who have these needs, and treat these personas as design tools, not evidence.

Link to the HAI principle. Each persona expresses a different need for control: Samuel wants explanations he can verify, Lucía wants reversible "what if" choices, and Nour wants editable assumptions. In design terms, the simulator should never lock in a plan, and should let users override any estimate.

Part 3: Data Source Description

Data source: the Synthetic University Course and Workload Dataset (SUCWD).

The Academic Decision Simulator will use a synthetic dataset generated by our team to represent a mid-sized university. It contains no real student records. It has four linked tables:

1. **Course catalog:** about 120 courses across 6 departments (for example Computer Science, Biology, Business, Mathematics, Humanities and Engineering), each with a code, credit value (mostly 3 credits), level, and the terms it is offered.
2. **Prerequisite and co-requisite tree:** roughly 200 links between courses, including chains up to four courses deep and a few “either/or” options, so the simulator can check whether a plan is legal and in what order courses must be taken.
3. **Workload estimates:** for each course, weekly hours broken down into lecture, reading, assignments, lab work and exam preparation, given as a range (low, typical, high) rather than a single number. These ranges are built from the common rule of thumb of about two hours of outside work for every hour in class, adjusted by course type (for example, labs and programming courses are heavier).
4. **Synthetic student-term records:** about 2,000 anonymous, generated “student semesters” showing which course combinations were taken, the total weekly hours, and a simple outcome (completed, struggled, withdrew), used to model how load combinations relate to burnout risk.

License. The dataset is created by the team, so we can release it under **Creative Commons Attribution 4.0 (CC BY 4.0)**, with the generation scripts under an MIT license. No third-party or personal data is included, so there are no privacy or consent constraints.

Why it fits the project. The simulator’s purpose is to let students test course combinations before committing, and that needs exactly three things: rules that decide which plans are possible (the prerequisite tree), a realistic estimate of weekly effort (the workload ranges), and a way to see how combined loads play out (the student-term records). Storing workload as ranges lets the simulator show uncertainty and let users replace estimates with their own, which supports our User Control and Agency principle. Using synthetic data also means we can deliberately include varied situations, such as heavy lab combinations and part-time loads, which public records rarely contain, while keeping real students’ information out of the project entirely. The main limitation is that synthetic estimates are not measured reality, and we will state this clearly in the interface.

Prompt Log:

	Tool	Exact prompt	What it gave	What we changed and why
3	Claude	Act as a Lead UX Researcher. I am designing an AI chatbot called the "Academic Decision Simulator" (Team: Ctrl+Z Burnout) to help university students plan their courses and avoid burnout. The mandatory HAI principle is "User Control and Agency". Please generate the deliverables for our "Users and Data" milestone based on the following strict constraints: Part 1: Fictional Personas Create exactly 3 fictional user personas (University Students). • Technical Levels: One must be "below typical", one "typical", and one "advanced". • AI Familiarity: At least 2 personas must be "Novice AI users". • Accessibility: At least 1 persona must have a clear accessibility need (e.g., ADHD, Dyslexia) that affects how they interact with schedules or text. • Diversity: They must have varied cultural or socio-economic backgrounds (e.g., international student, working student, commuter).	Generated 3 diverse personas (Samuel, Lucía, Nour), a detailed bias-check table, and a description of the synthetic SUCWD dataset.	**Change:** We refined the accessibility needs to directly influence our UI. For example, we ensured ADHD and Dyslexia were framed as interaction preferences (e.g., needing step-by-step choices) rather than deficits. We also finalized the dataset size to 120 courses and 2,000 records. **Why:** To ensure the personas explicitly drive our "User Control" design and to fully meet the rubric's synthetic data constraints.

Tool	Exact prompt	What it gave	What we changed and why
	• *Format for each:* Name, Background, Tech Level, AI Familiarity, Accessibility Need, Goals, Pain Points. Part 2: Bias-Check Notes Provide a section detailing the bias-check process. Explain what stereotypes or exclusions were reviewed (e.g., gender roles in tech, framing neurodivergence as a burden, assuming all students have unlimited free time), and what was actively changed in the personas to avoid them. Part 3: Data Source Description Write a half-page description of the data source this AI simulator will use. The data must be described as *public or synthetic* (e.g., a synthetic university course catalog with prerequisite trees and historical workload hour estimates). Explain exactly what the data is, its approximate size/scope, its license (e.g., open-source/creative commons), and logically justify why this specific data fits the project's goal of simulating academic workloads.		